

Clinical Risk Assessment Report: Patient 5

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.9% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.54x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 3.6%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

- Large Zone Emphasis (CT) (GLZLM_LGZE.CT)

Value: -0.3 [Bottom 24%] (+1.5%)

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.
- Squamous Cell Carcinoma Antigen (SCC_Ag)

Value: 0.5 [Bottom 10%] (+1.3%)

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

- Small Zone Emphasis (PET) (GLZLM_SZE.PET)

Value: -0.52 [Avg (27th %ile)] (-3.6%)

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.
- Patient Age (Age)

Value: 69.0 [Avg (73th %ile)] (-1.5%)

Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.
- 75th Percentile Tumor Density (CONVENTIONAL_HUQ3)

Value: 0.37 [Avg (54th %ile)] (-1.1%)

Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.
- Maximum Tumor Density (CONVENTIONAL_HUmax)

Value: -0.13 [Avg (40th %ile)] (-0.9%)

Reflects the most dense solid components or calcifications within the primary tumor lesion.
- 25th Percentile Metabolic Activity (CONVENTIONAL_SUVbwQ1)

Value: -0.59 [Avg (32th %ile)] (-0.7%)

Reflects the baseline metabolic activity and glucose consumption of the tumor tissue.
- Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET)

Value: -0.46 [Avg (36th %ile)] (-0.6%)

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Machine Learning Feature Attribution (SHAP Decision Path)

